

TECTONIC HAZARDS – REVISION MAT



Tectonic hazards are natural events caused by the movement of the Earth's plates. They **shape** our planet and can affect people and the environment in many ways.

1. STRUCTURE OF THE EARTH

The Earth has several layers. The four main layers are:

1 CRUST

- Thin, outer rocky layer.
- Broken into tectonic plates.

2 MANTLE

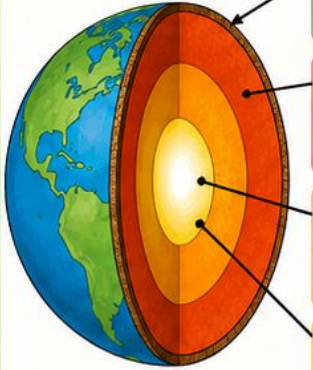
- Thick layer of hot rock (magma).
- Heat from the core causes movement.

3 OUTER CORE

- Liquid iron and nickel.
- Very hot.
- Creates Earth's magnetic field.

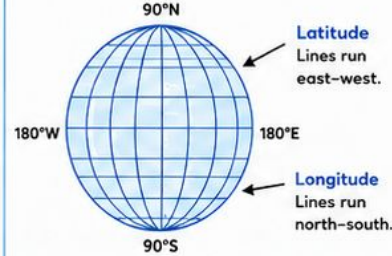
4 INNER CORE

- Solid iron and nickel.
- Extremely hot.
- Centre of the Earth.



2. LATITUDE & LONGITUDE

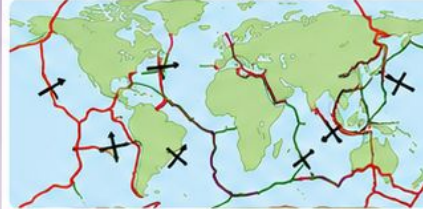
We use a grid of imaginary lines to find places on Earth.



- **Latitude:** how far north or south of the Equator (0° at Equator)
- **Longitude:** how far east or west of the Prime Meridian (0° at Prime Meridian)

3. TECTONIC PLATES

The Earth's outer layer is broken into large pieces called **tectonic plates**.



TYPES OF PLATE BOUNDARIES

- Destructive (plates move together) – one plate forced under another.
- Constructive (plates move apart) – magma rises to the surface.
- Conservative (plates slide past each other).
- Direction of plate movement

4. VOLCANOES

Volcanoes form when magma rises to the surface. There are two main types.

SHIELD VOLCANO

- Wide, gentle slopes
- Runny basaltic lava
- Less explosive eruptions
- Example: Mauna Loa (Hawaii)



COMPOSITE VOLCANO

- Tall, steep sides
- Sticky, silica-rich lava
- More explosive eruptions
- Example: Mount Fuji (Japan)



5. WHY STICKY MAGMA CAN CAUSE EXPLOSIVE ERUPTIONS

Sticky magma is thick and traps gas bubbles.



Gas cannot escape easily, so pressure builds up.



When the pressure becomes too high, the magma explodes out of the volcano.



This causes an **EXPLOSIVE ERUPTION**.

6. VOLCANIC HAZARDS

Volcanic eruptions can create many hazards that can affect people and the environment.

	Ash	Can cause breathing problems, reduce visibility, damage buildings and disrupt transport.
	Lava	Very hot and destroys buildings, roads, crops and natural habitats.
	Pyroclastic flows	Fast-moving, hot clouds of ash, gas and rock. Can destroy everything in their path.
	Volcanic bombs	Large rocks thrown out of the volcano. Can injure or kill people and damage buildings.
	Lahars	Mudflows caused by ash, rain or melted snow. Can bury villages and destroy infrastructure.

7. IMPACTS OF VOLCANIC ERUPTIONS

IMPACTS ON PEOPLE

- Deaths and injuries
- People may need to evacuate
- Homes and infrastructure destroyed
- Loss of jobs and income
- Ash can cause health problems
- Transport and services disrupted
- Farmland and crops can be destroyed



IMPACTS ON THE ENVIRONMENT

- Habitats destroyed
- Plants and animals killed or displaced
- Landscapes can change
- Ash and gases can pollute air and water
- Lava can cover large areas of land



8. KEY WORDS

Magma

Molten rock below the Earth's surface.

Lava

Magma that erupts onto the Earth's surface.

Epicentre

The point on the Earth's surface above an earthquake.

Tectonic plate

A large section of the Earth's crust that moves.

Hazard

A natural event that can cause harm.



Review your notes, diagrams and key vocabulary.



Practise using maps, grids and keys to find and describe places.



Think about how different processes are connected and why they happen.



Use examples to support your answers whenever you can.



Explain your ideas clearly and use geographical words accurately.